

```
1
2  cd "F:\ECON 408\Practice do
   file\Assignments\Time-Series-Assignment\Assugnment-45\Do
   file"
3
4  clear all
5  set more off
6  set seed 10
7
8  drop _all
9  set obs 100
10 gen t = _n
11 tsset t
12
13 set scheme white_tableau
14 graph set window fontface "CMU Serif"
15 graph set eps fontface "CMU Serif"
16 graph set svg fontface "CMU Serif"
17
18
19 * Generating 500 Random walk process with a loop
20
21 forvalues i = 1/500 {
22     gen double u_`i' = rnormal()
23     gen double y_`i' = sum(u_`i')
24 }
25
26
27 * Creating columns and appending 400 rows to store 500
   regression results
28
29 gen t_stat = .
30 gen r2 = .
31 tsappend , add(400)
32
33 forvalues i = 2/500 {
34     quietly {
35         reg y_1 y_`i'
36         replace t_stat = _b[y_`i'] / _se[y_`i'] in `i'
37         replace r2 = e(r2) in `i'
38     }
39 }
```

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40
41
42 * Test Statistics and R2 Squared analysis
43
44 gen reject = (abs(t_stat) > 1.96) if (t_stat !=.)
45
46 display "Target False Rejection Rate: 0.05 (5%)"
47 sum reject
48
49 display "Average R-squared of completely independent
variables:"
50 sum r2
51
52
53 * Visualizations
54
55 histogram r2, ///
56     width(0.05) ///
57     fcolor(ebblue%70) ///
58     lcolor(white) lwidth(vthin) ///
59     normal ///
60     normopts(lcolor(pink)) ///
61     title("Spurious Regression: R-squared Distribution",
color(black) size(medlarge)) ///
62     xtitle("Estimated R-squared") ///
63     ytitle("Density") name(histogram_hw5 , replace)
64
65 graph export histogram_hw5.png, name(histogram_hw5) replace
66
67
68 twoway ///
69     (kdensity t_stat, recast(area) fcolor(ebblue%70)
lcolor(white) lwidth(medthick)) ///
70     (function y = normalden(x), range(-10 10) lcolor(red)
lwidth(medthick) lpattern(dash)), ///
71     title("Spurious Regression t-statistic Distribution")
///
72     subtitle("500 Independent Random Walks (T=100)") ///
73     xtitle("OLS t-statistic for Beta") xlabel(,nogrid) ///
74     ytitle("Density") ylabel(,nogrid) ///
75     legend(order(1 "Simulated t-stat" 2 "Standard Normal")
col(1) pos(2) ring(0)) ///

```

```
76      xline(-1.96 1.96, lcolor(black) lpattern(dot) lwidth(  
      thick)) name(kdensity_hw5,replace)  
77  
78  graph export kdensity_hw5.png , name(kdensity_hw5) replace  
79  
80  
81  
82
```